Aster Memory Lab - definitions

A virtual page is a fixed-size block of virtual addresses. A physical frame is a fixed-size block of physical memory. In this lab both have size 4096 bytes. The page table maps virtual page numbers to physical frame numbers. The offset within a page is preserved during address translation.

Aster Memory Lab - cache rules

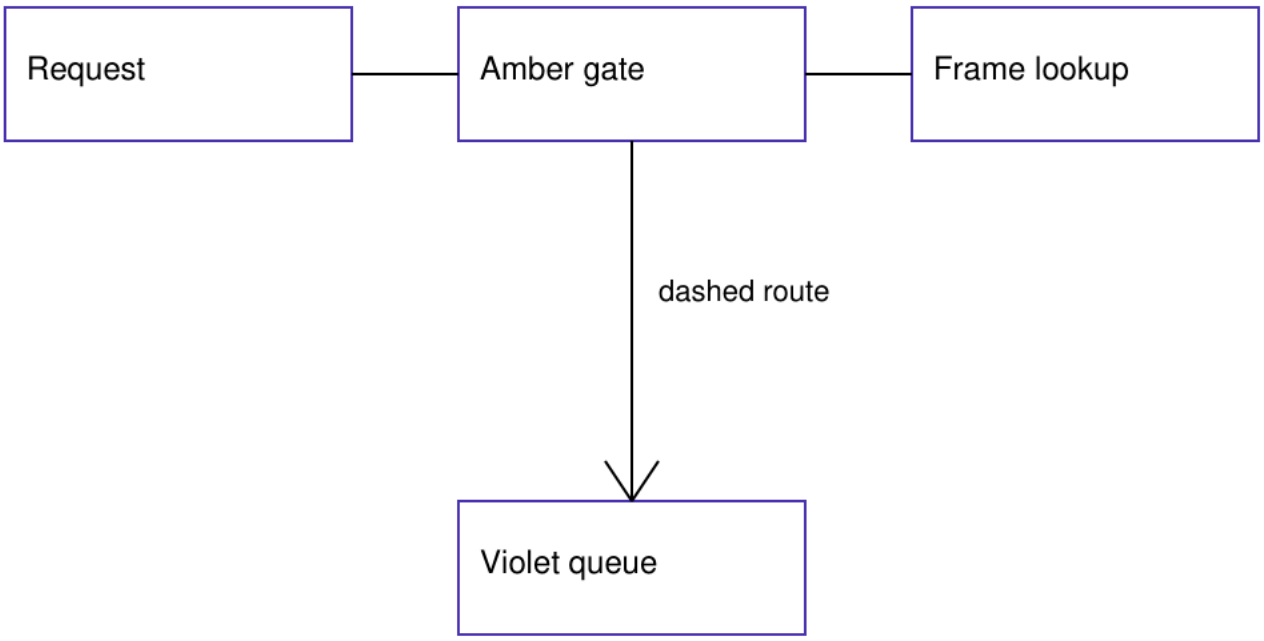
The translation lookaside buffer (TLB) caches recent virtual-page to physical-frame translations. A TLB miss requires consulting the page table; it does not by itself imply that the page is absent from physical memory. A page fault occurs when the referenced virtual page is not resident. The page table entry has a present bit that indicates residency. A present bit of zero triggers the page-fault handler.

Aster Memory Lab - measured trace

The following table reports this lab's frozen trace. Hit fraction is hits divided by lookups. These are trace measurements, not universal performance guarantees.

Trace	TLB lookups	TLB hits	Page faults
Birch	100	80	3
Cedar	200	150	5
Elm	120	90	2

Aster routing figure (read the embedded diagram)



Dashed route leaves Amber gate and enters Violet queue.